



Instruction Booklet

Logic Puzzle Open

April 4, 2026 • 10 AM – 4:30 PM • Room 32-141

10:00 AM – 10:50 AM	Round 1: “Pop Culture” <i>Sudoku, Numberlink, Domino Hunt, TomTom, Battleships</i>
11:00 AM – 11:50 AM	Round 2: Heads Up <i>Fireflies, Skyscrapers, Clouds, Moon or Sun, Spiral Galaxies</i>
12:00 PM – 12:50 PM	Lunch Break
1:00 PM – 1:50 PM	Round 3: Powerhouses <i>Tapa, Kurotto, Yajilin, Easy As, Balance Loop</i>
2:00 PM – 2:50 PM	Round 4: Grab Bag <i>Hanami, Dominion, Stitches, Voxas, Nurikabe Loop</i>
3:10 PM – 4:00 PM	Round 5: Team Round
4:00 PM – 4:30 PM	Awards

Competition Rules

You can have pens, pencils, sharpeners, erasers, rulers, blank papers, and a copy of this instruction booklet. We'll have extras for all of these, but don't rely on them. We won't print a copy of the full booklet for everyone, but we will print a copy of the last two pages for everyone.

Puzzle booklets will have the rules but not example puzzles.

Notation

Each section of this instruction booklet has five puzzle types. Each puzzle type comes with the rules, an example puzzle, and the solution to the example.

We'll use standard notation for example solutions, but **you can use whatever notation you want**, as long as you are consistent throughout a solution, and it is clear to us what you mean.

If multiple marks are in the same cell, we'll consider the largest such mark. This is to let you make small notes on the corners or edges of the cells without having to erase them. If we're not sure what the intended final mark is, we'll consider that cell blank.

Scoring

Each individual round is worth **1000** points and has 15 puzzles. Puzzles that took testsolvers longer to solve are worth more points.

You'll be given full points for a correctly solved puzzle. You may get 80% partial credit for a solution with a minor error (e.g. 1 missing shaded cell), at our discretion.

If you finish early, say "Finished", and a proctor will collect your booklet. If all puzzles are solved correctly or with at most two minor errors, you'll receive **20** bonus points per saved minute. After finishing, you can leave the room, but make sure not to disturb anyone.

Divisions and Prizes

We have two divisions: Advanced and Beginner. Each division has different puzzles: for each genre, the Advanced division has a harder puzzle, while the Beginner division has an easier puzzle.

We have prizes for the top 5 solvers in each division, plus the next 5 MIT students. We might also have *special* awards, at our discretion.

The team round won't have prizes (other than bragging rights); just solve for fun!

Credits

Most instructions are copied from **Eric Fox's** puzzle rules document.

Puzzles were constructed by **Barbitos**, **djmathman**, **Rever**, and **yosh**.

Testsolvers were **CJ**, **lumia**, **Walker**, and **Wei-hwa**.

Tools used include **Penpa+**, **puzz.link**, and **Affinity Publisher**.

Round 1: “Pop Culture”

These puzzle types have all broken into the mainstream in some way.

1.1–1.3 Sudoku

Insert a number from 1 to 9 (1 to 6 in the first puzzle) so that no number repeats in any row, column, or box.

	1			6	
3					4
		5			
			6		
2					5
	4			3	

4	1	2	5	6	3
3	5	6	1	2	4
6	2	5	3	4	1
1	3	4	6	5	2
2	6	3	4	1	5
5	4	1	2	3	6

1.4–1.6 Numberlink (a.k.a. Flow Free®)

Draw non-intersecting paths through the centers of some cells, each connecting one clue to its equal counterpart.

Note: In these puzzles, every cell will be used by some path. Feel free to use this to your advantage.

1					
2					5
3			3		
		4			
	5	2			
				1	
					4

1					
2					5
3			3		
		4			
	5	2			
				1	
					4

1.7–1.9 Domino Hunt

Divide the grid into dominoes so that each domino in the indicated set is used exactly as many times as it appears.

The orientation of the digits does not matter.

2	2	2	1
0	1	1	2
0	0	0	1

0 0

0 1 1 1

0 2 1 2 2 2

2	2	2	1
0	1	1	2
0	0	0	1

0 0

0 1 1 1

0 2 1 2 2 2

1.10–1.12 TomTom

Place a number from 1 to N into each cell so that each row and column contains every number from that range with no repeats, where N is the side length of the grid.

A clue represents the value obtained by applying an operation iteratively on the numbers in the region the clue is in.

If no operation is given, it may be any of +, -, ×, or ÷.

Subtraction and division in regions with more than two numbers are handled by taking the largest number and subtracting/dividing all the others.

9+		2-	
	12		
2÷			24×

9+	3	2	2-	1	4
	4	12	3	2	1
2÷	1	4	3	24×	2
	2	1	4	3	

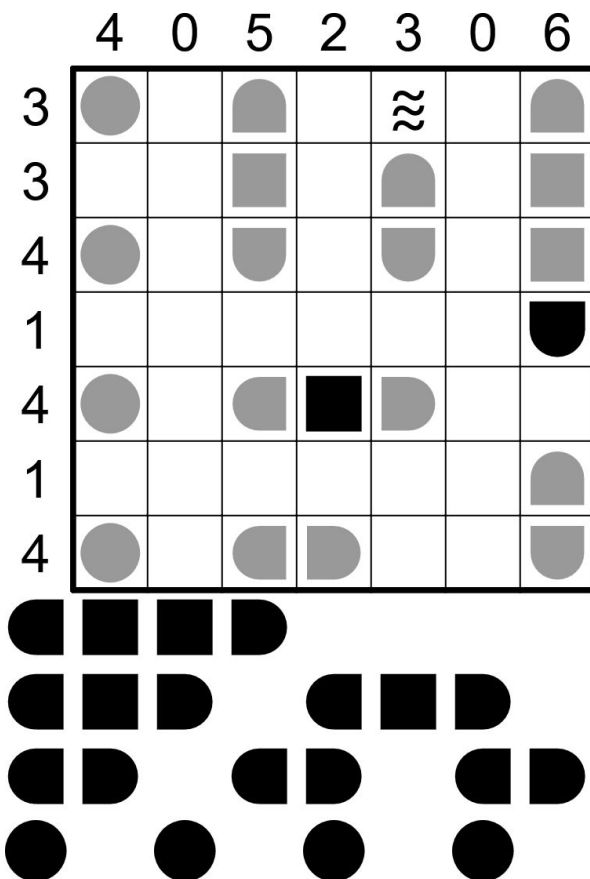
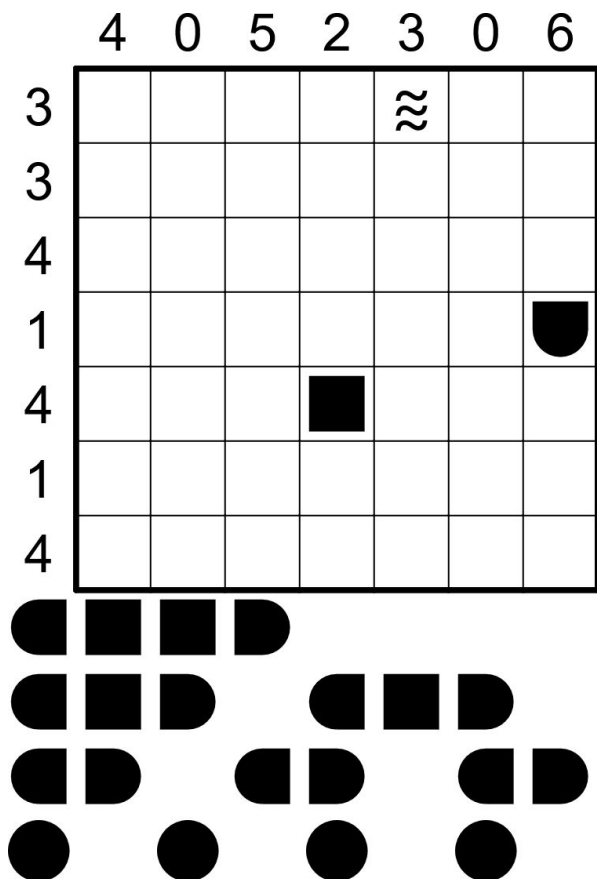
1.13–1.15 Battleships

Place the given fleet of ships into the grid so that no two ships are touching, not even diagonally. Rotating ships is permitted.

A clue outside the grid indicates the number of cells in the corresponding row or column that are occupied by ships.

Cells with waves cannot be occupied by a ship.

A given ship segment must be used as the part of a ship that its shape represents.



Round 2: Heads Up

These puzzle types are all themed around things you can see in the sky.

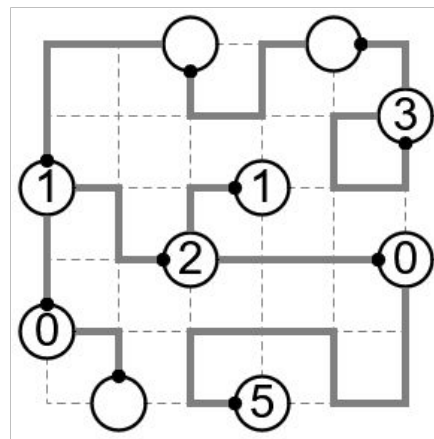
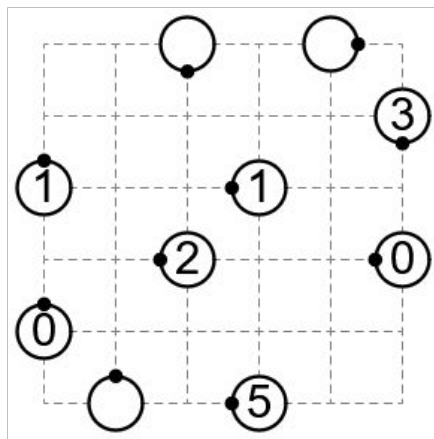
2.1–2.3 Fireflies (a.k.a. Hotaru Beam)

For each circle in the grid, draw a path starting from its black dot, moving along the gridlines, and ending at a circle (including potentially the same one it started at), but not on the side containing another black dot.

Paths may not cross themselves, each other, or pass through circles.

A number in a circle represents how many turns the path exiting from its dot will make.

All circles must be joined by paths to form one connected network.



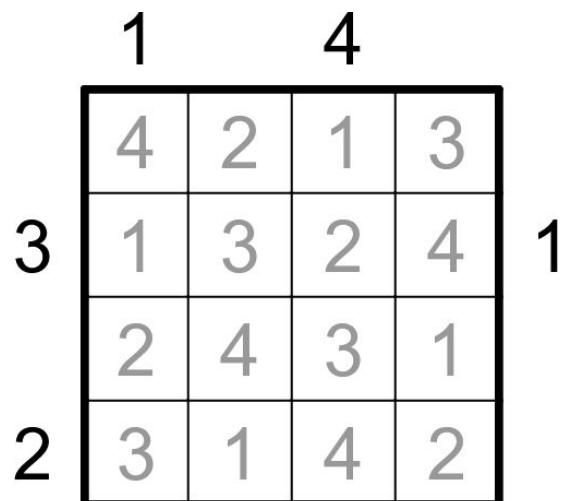
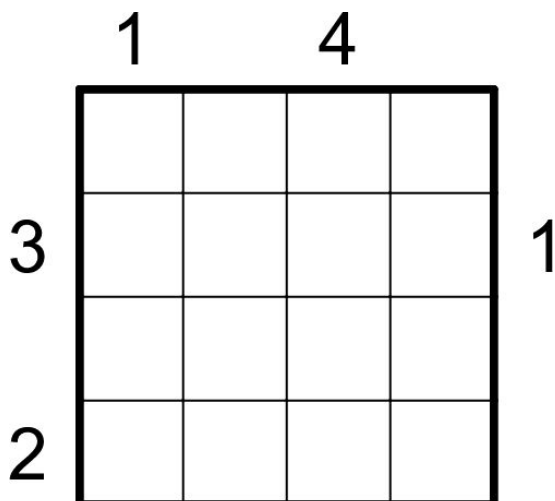
2.4–2.6 Skyscrapers

Place a number from 1 to N into each cell so that each row and column contains every number from that range with no repeats, where N is the side length of the grid.

Each number represents a skyscraper of its respective height.

The numbers outside the grid indicate how many skyscrapers can be seen in the respective row or column from the respective direction; shorter skyscrapers are hidden behind taller ones.

Some numbers may already be filled in for you.

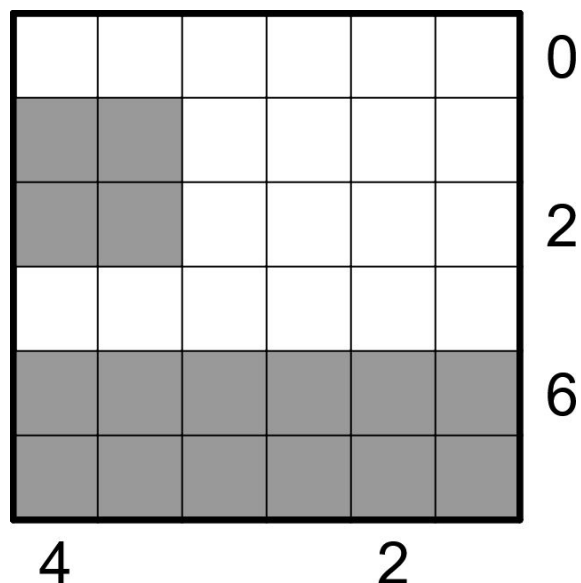
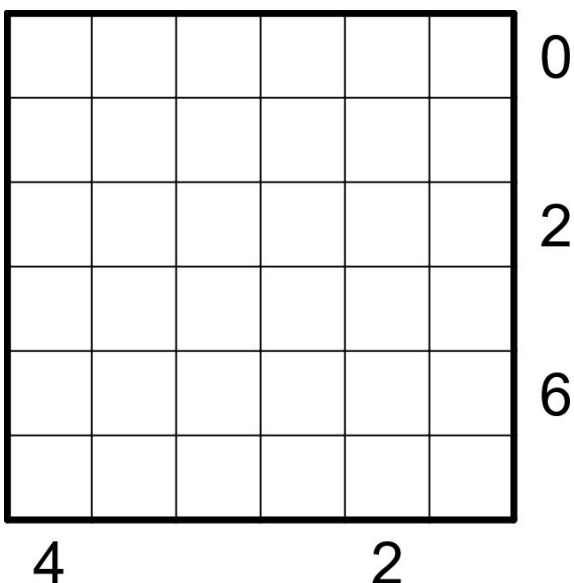


2.7–2.9 Clouds

Shade some cells so that each orthogonally connected area of shaded cells is in the shape of a rectangle at least two cells wide and at least two cells tall.

Rectangles may not touch one another, not even diagonally.

A clue outside the grid indicates the number of shaded cells in the corresponding row or column.



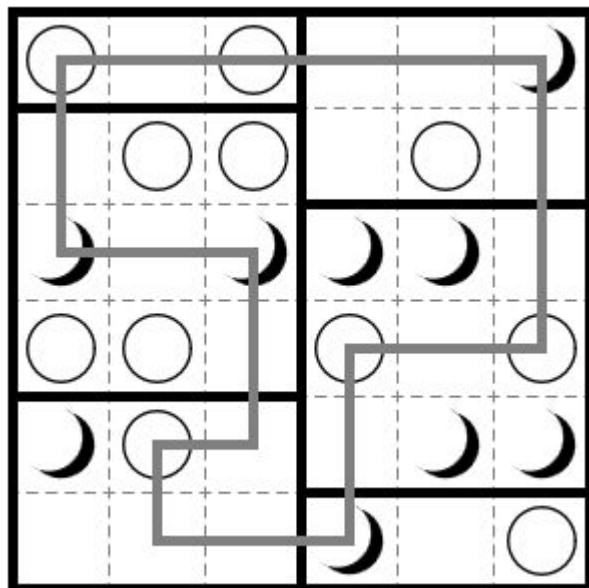
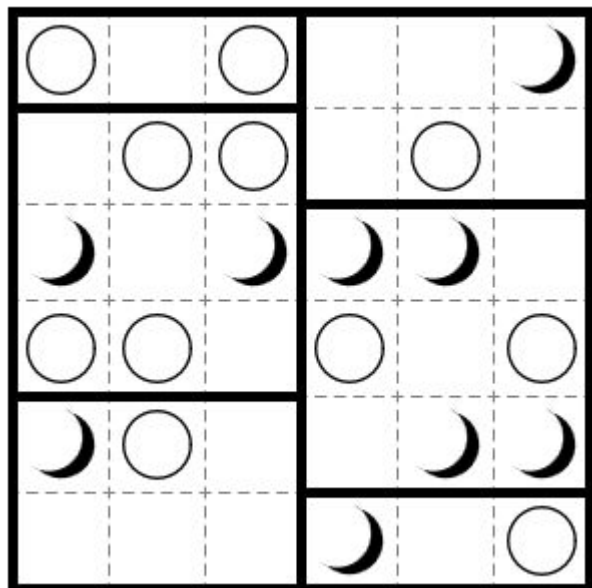
2.10–2.12 Moon or Sun

Draw a non-intersecting loop through the centers of some cells which passes through each region exactly once.

Within a region, the loop must pass through all moons and no suns, or all suns and no moons.

A region containing only suns or only moons must have its clues passed through.

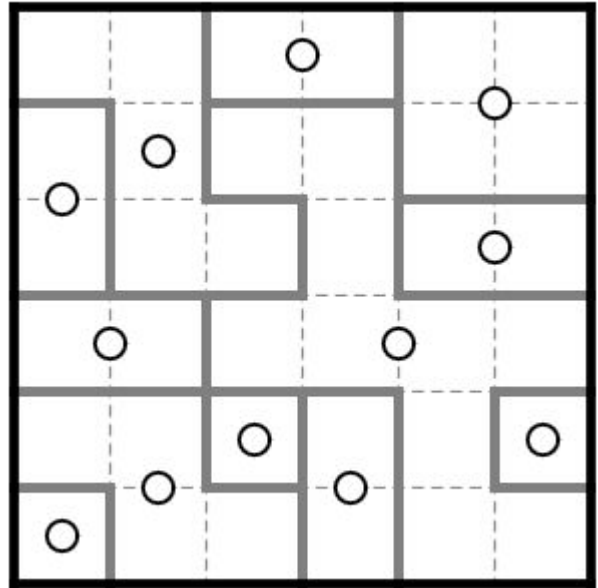
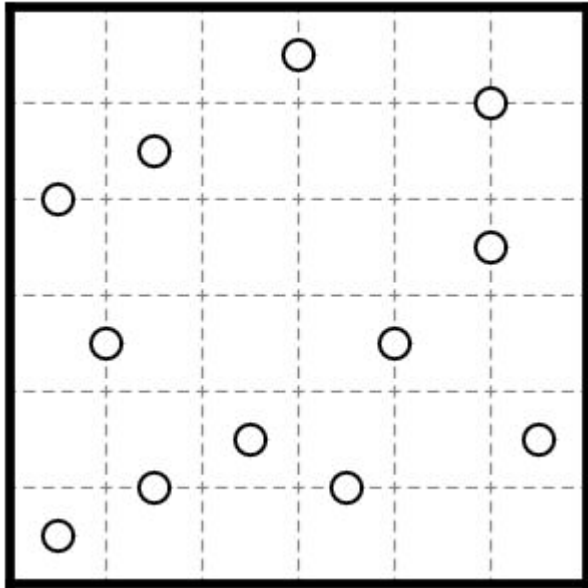
The loop may not pass through the same type of clue in two consecutively used regions.



2.13–2.15 Spiral Galaxies

Divide the grid into regions of orthogonally connected cells.

Each region must contain exactly one circle and have 180° rotational symmetry around it.



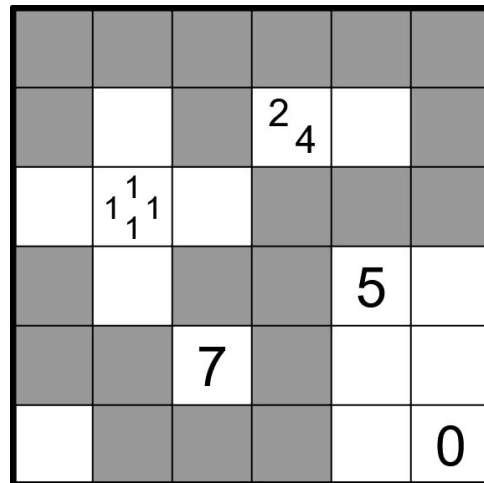
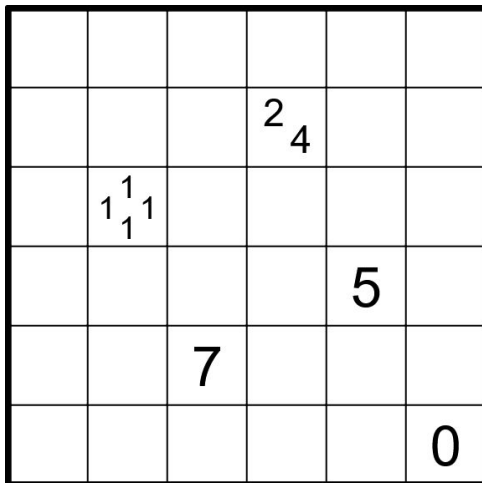
Round 3: Powerhouses

These puzzle types are ubiquitous within logic puzzle circles.

3.1–3.3 Tapa

Shade some cells so that all shaded cells form one orthogonally connected area and no 2x2 region is entirely shaded.

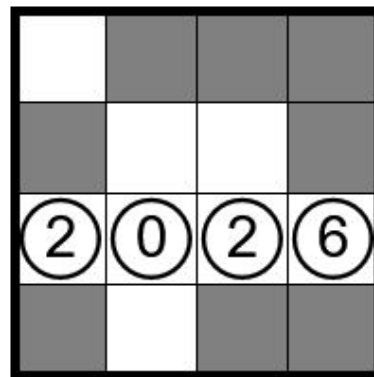
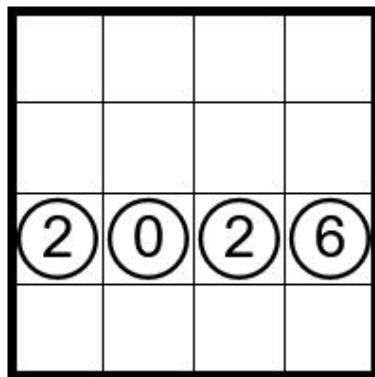
Clues cannot be shaded, and represent the lengths of the blocks of consecutive shaded cells in the (up to) eight cells surrounding the clue.



3.4–3.6 Kurotto

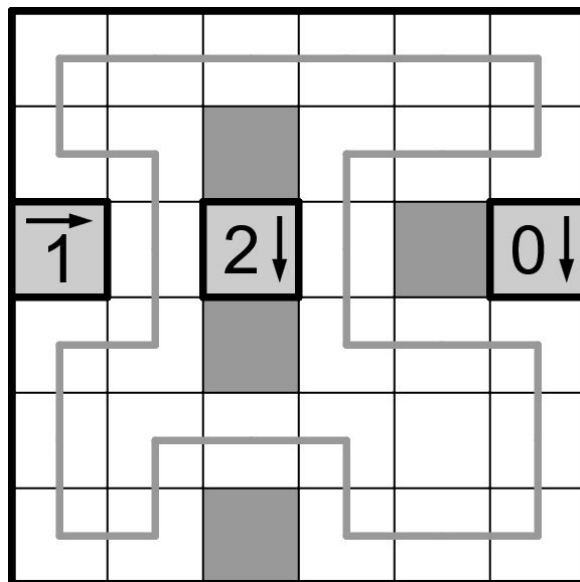
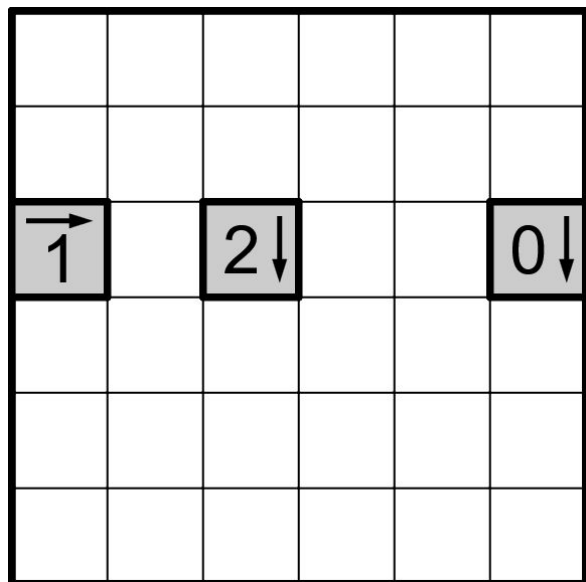
Shade some cells so that clues represent the total size of the orthogonally connected areas of shaded cells that share an edge with the clue.

Clued cells cannot be shaded.



3.7–3.9 Yajilin

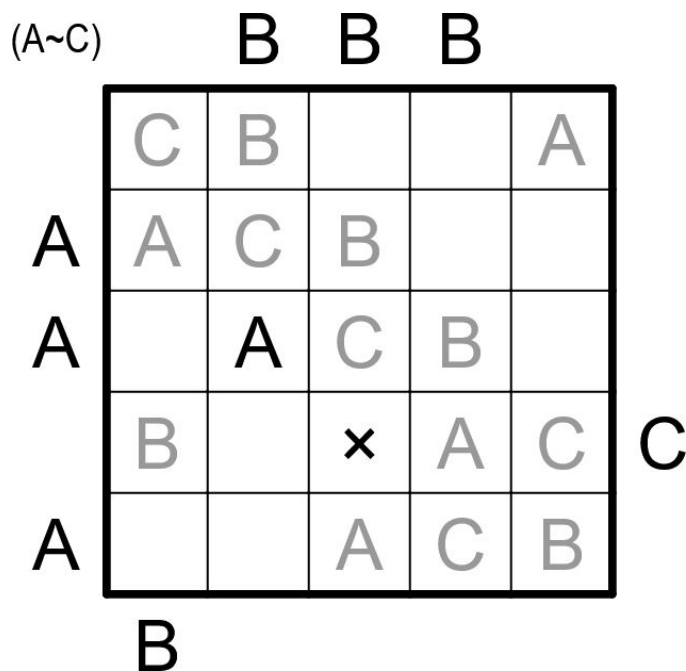
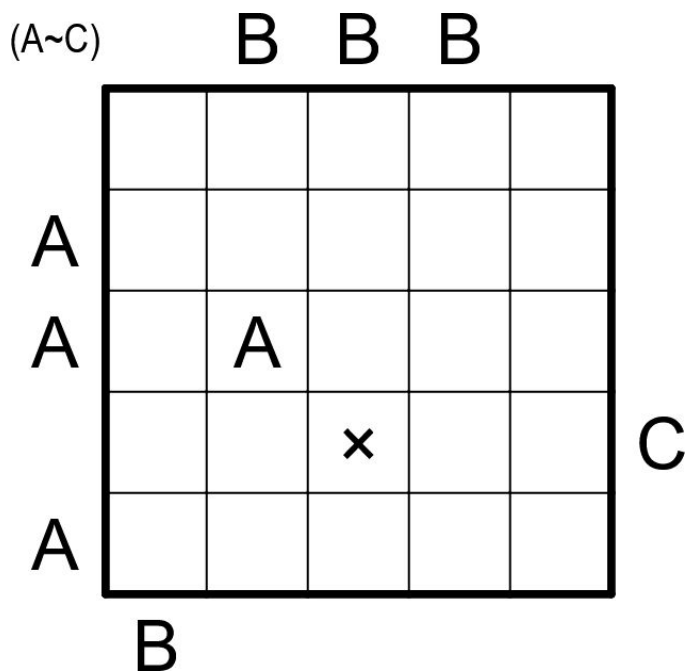
Shade some cells so that no two shaded cells are orthogonally adjacent, and draw a non-intersecting loop through the centers of all the remaining empty cells. Clues cannot be shaded, and represent the number of shaded cells in a straight line in the indicated direction.



3.10–3.12 Easy As

Place letters from the range given outside the grid into some cells so that each row and column contains each letter once.

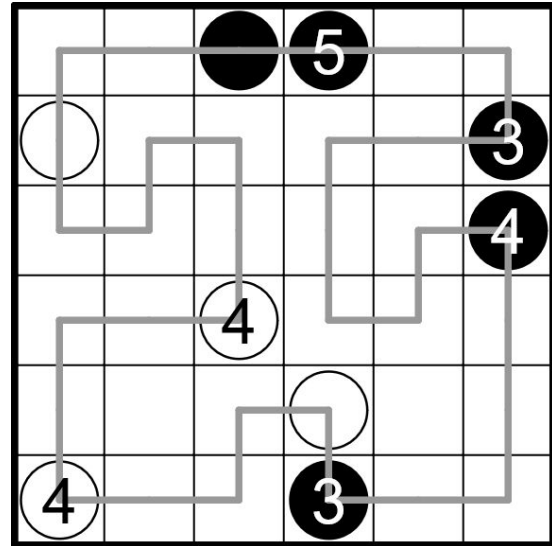
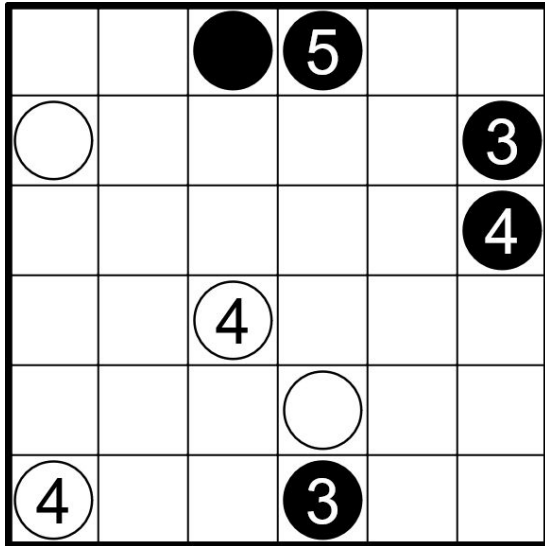
A clue outside the grid represents the first letter seen in the corresponding row or column from that direction.



3.13–3.15 Balance Loop

Draw a non-intersecting loop through the centers of some cells that passes through every circle. The straight line segments coming out of a white circle must have equal length, while the straight line segments coming out of a black circle must have different lengths.

A clue in a circle represents the sum of the lengths of these two line segments.



Round 4: Grab Bag

These puzzle types don't follow a particular theme; we just wanted to use them.

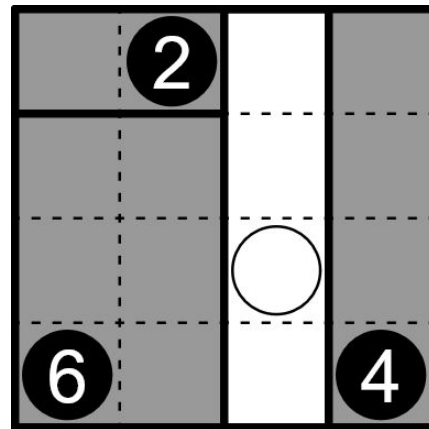
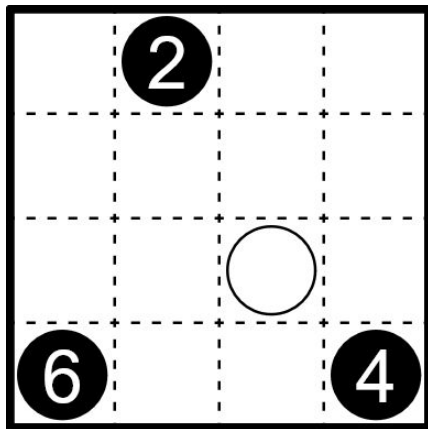
4.1–4.3 Hanami

Draw some rectangles in the grid.

Rectangles cannot overlap each other or white circles.

Every rectangle must contain exactly one black circle, whose number represents the area of the rectangle.

The cells not used by any rectangle must form a single connected area.

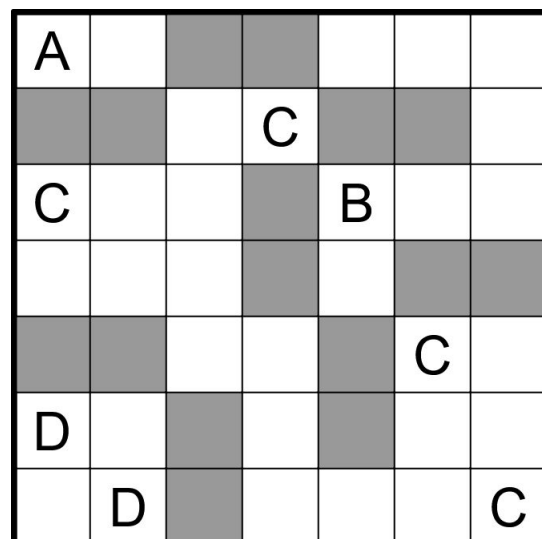
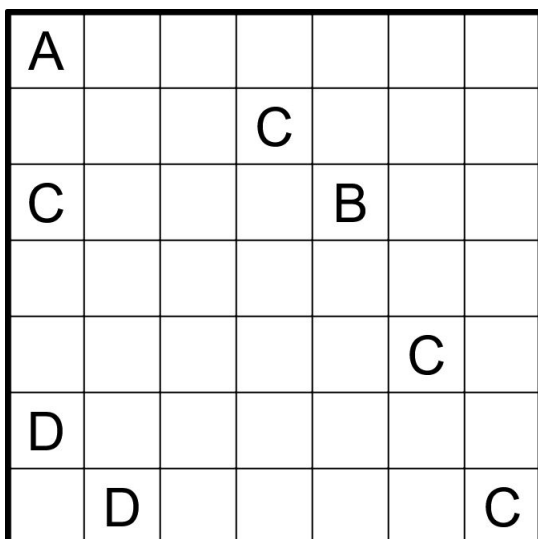


4.4–4.6 Dominion

Shade some dominoes of cells to divide the grid into unshaded areas.

Shaded dominoes may not touch orthogonally. Clues cannot be shaded.

Each orthogonally connected area of unshaded cells contains exactly one type of clue, and all instances of it.



4.7–4.9 Stitches

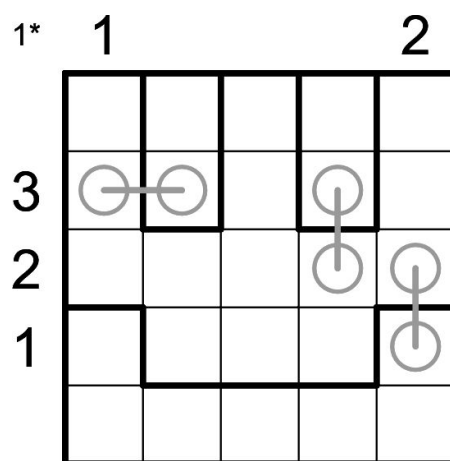
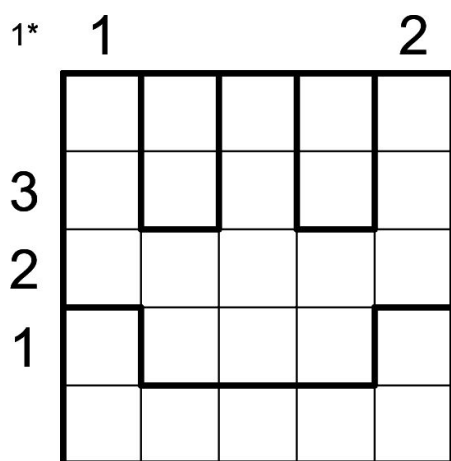
Draw some lines of length 1 ("stitches") in the grid.

Each cell contains at most one stitch.

All stitches must cross a region border, "connecting" the two regions.

Every pair of regions sharing at least one edge must have exactly N stitches connecting them, where N is specified next to the grid.

Clues outside the grid indicate how many cells are used by stitches in the corresponding row/column.



4.10–4.12 Voxas

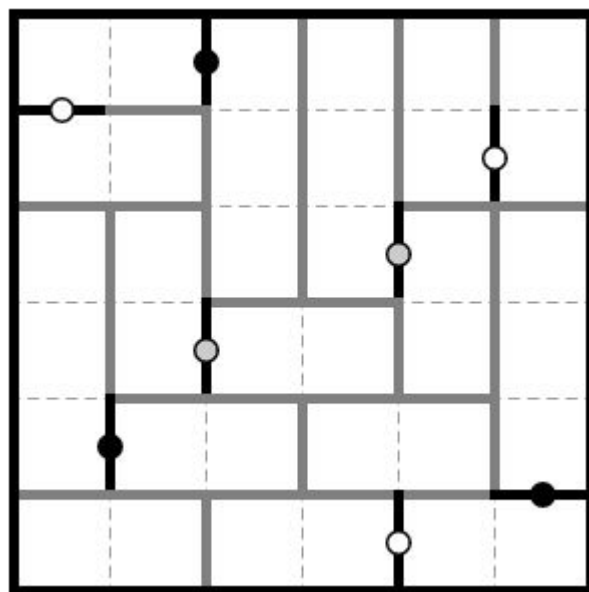
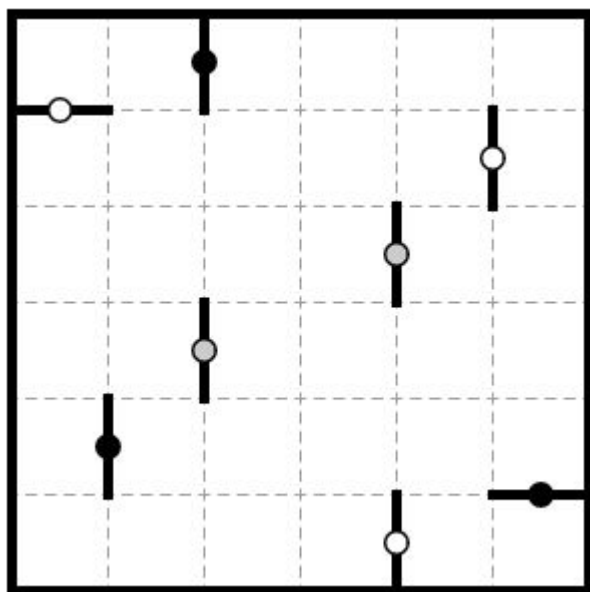
Divide the grid into 1x2 and 1x3 regions.

Borders must separate two different regions.

Borders with white dots separate regions with the same size and orientation.

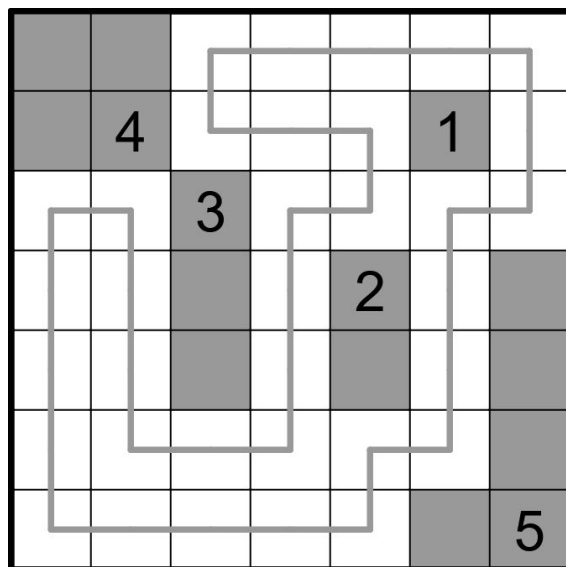
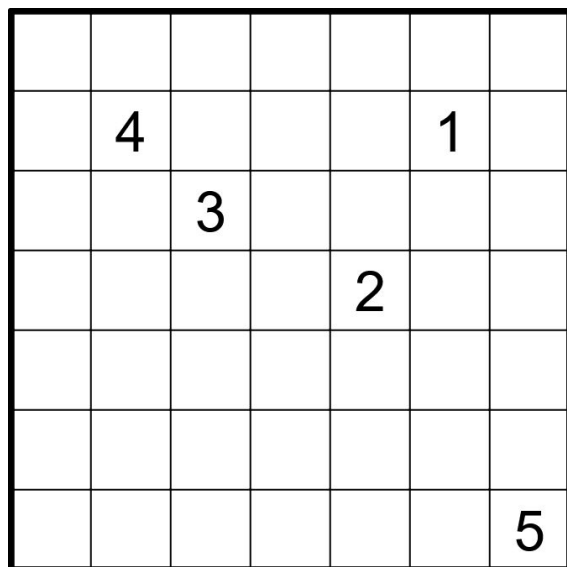
Borders with black dots separate regions with neither the same size nor the same orientation.

Borders with grey dots separate regions with either the same size or the same orientation, but not both.



4.13–4.15 Nurikabe Loop

Draw a non-intersecting loop through the centers of some empty cells. Every orthogonally connected area of cells not visited by the loop contains exactly one clue, the value of which represents the size of the area.



Round 5: Team Round

5.01 - 5.?? Kurotto with Instructionless Variants

Shade some cells so that circled clues, which cannot be shaded, represent the total size of the orthogonally connected areas of shaded cells that share an edge with the clue.

Note: It is allowed for a group of shaded cells to not touch any circled clues.

There are additional clues and variants; it is up to the solving team to determine what they mean.

Team Round Format

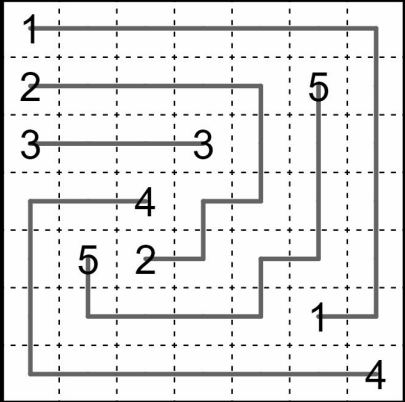
The team round will feature an assortment of different Kurotto puzzles with various sizes.

Members in a team should work together to discover the meanings of the new clues and variants, which will show up in multiple puzzles for teams to parallelize.

Sudoku

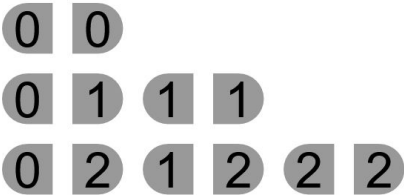
4	1	2	5	6	3
3	5	6	1	2	4
6	2	5	3	4	1
1	3	4	6	5	2
2	6	3	4	1	5
5	4	1	2	3	6

Numberlink



Domino Hunt

2	2	2	1
0	1	1	2
0	0	0	1



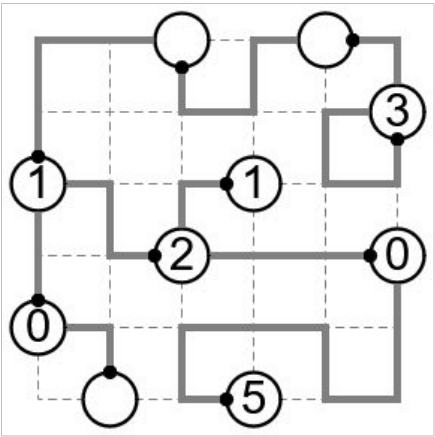
TomTom

9+ 3	2	2- 1	4
4	12 3	2	1
2÷ 1	4	3	24x 2
2	1	4	3

Battleships

	4	0	5	2	3	0	6
3	●		◐		≈		◐
3			■		◐		■
4	●		◐		◐		■
1							◐
4	●		◐	■	◐		
1							◐
4	●		◐	◐			◐

Fireflies



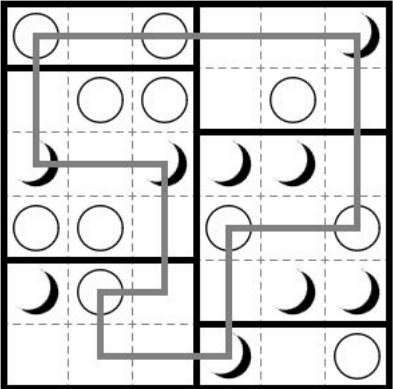
Skyscrapers

	1		4	
4	2	1	3	
1	3	2	4	
2	4	3	1	
3	1	4	2	

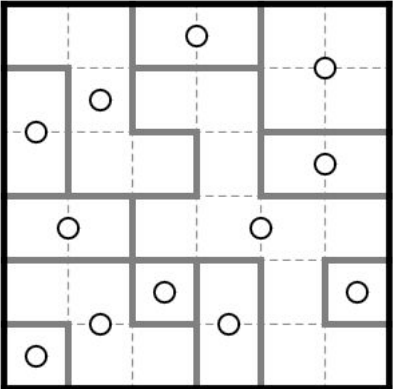
Clouds

						0
						2
						6
4					2	

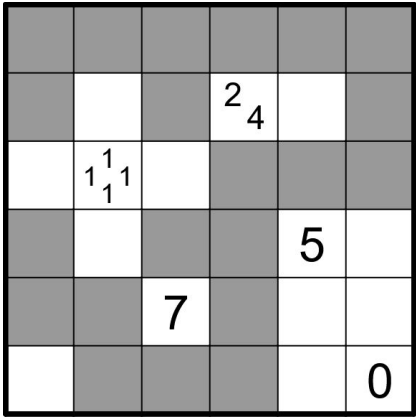
Moon or Sun



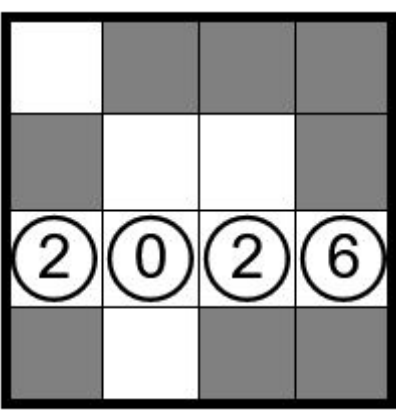
Spiral Galaxies



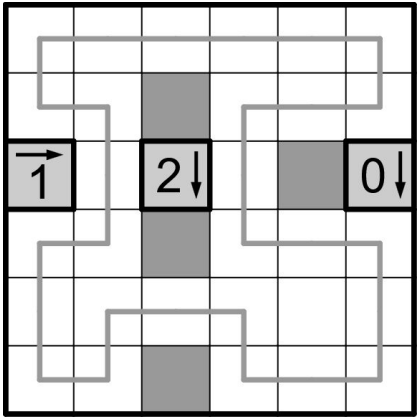
Tapa



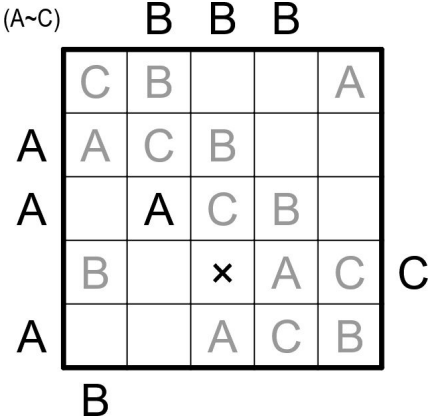
Kurotto



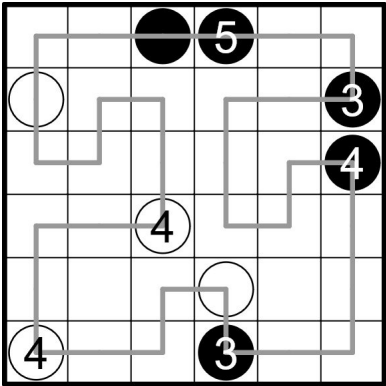
Yajilin



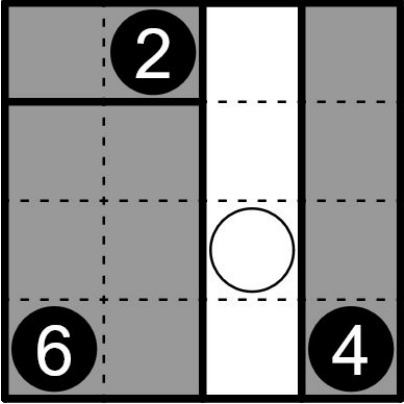
Easy As



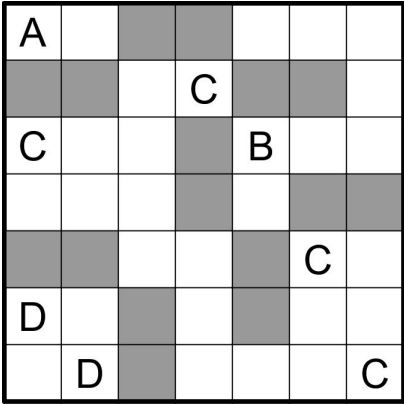
Balance Loop



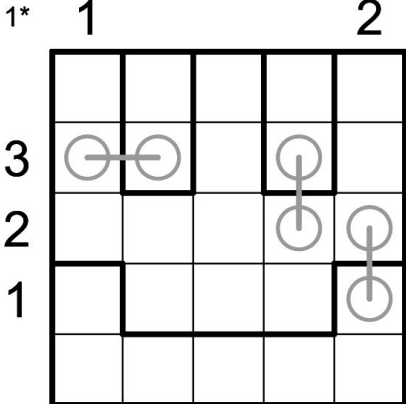
Hanami



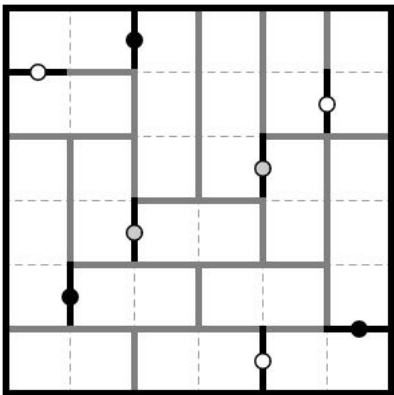
Dominion



Stitches



Voxas



Nurikabe Loop

